Just as the components of a vector change when we change the basis of the vector space, the components of a tensor also change under such a transformation. Each type of tensor comes equipped with a *transformation law* that details how the components of the tensor respond to a change of basis. The components of a vector can respond in two distinct ways to a change of basis (see covariance and contravariance of vectors), where the new basis vectors $\hat{\mathbf{e}}_i$ are expressed in terms of the old basis vectors $\mathbf{e}_j$ as,

$$\hat{\mathbf{e}}_i = \sum_{j=1}^{n} \mathbf{e}_j R_i^j = \mathbf{e}_j R_i^j.$$

Here $R^j{}_i$ are the entries of the change of basis matrix, and in the rightmost expression the summation sign was suppressed: this is the Einstein summation convention, which will be used throughout this article.[1] The components v^i of a column vector $\mathbf{v}$ transform with the inverse of the matrix R,

$$\hat{v}^i = (R^{-1})^i_j v^j,$$

where the hat denotes the components in the new basis. This is called a *contravariant* transformation law, because the vector components transform by the *inverse* of the change of basis. In contrast, the components, w_i, of a covector (or row vector), $\mathbf{w}$, transform with the matrix R itself,

$$\hat{w}_i = w_j R_i^j.$$

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[1] The Einstein summation convention, in brief, requires the sum to be taken over all values of the index whenever the same symbol appears as a subscript and superscript in the same term. For example, under this convention $B_i C^i = B_1 C^1 + B_2 C^2 + \cdots B_n C^n$